

Build an Artificial Neural Network by implementing the Backpropagation algorithm and test the same using appropriate data sets.

Aim

To build an **Artificial Neural Network (ANN)** by implementing the **Backpropagation algorithm** and test its performance using an appropriate dataset.

Apparatus / Requirements

- Google Colab

Procedure

1. Open Google Colab and import the required Python libraries.
2. Load the Breast Cancer Wisconsin dataset.
3. Split the dataset into training and testing sets.
4. Normalize the input features using StandardScaler.
5. Build an Artificial Neural Network (ANN) with input, hidden, and output layers.
6. Compile the model using the Adam optimizer and binary cross-entropy loss function.
7. Train the ANN using the Backpropagation algorithm.
8. Test the trained model using the test dataset.
9. Predict the output for the test samples.
10. Calculate the model accuracy and evaluate its performance.

Code :

```
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
import tensorflow as tf
```

```
data = load_breast_cancer()

X, y = data.data, data.target


X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)


sc = StandardScaler()
X_train = sc.fit_transform(X_train)
X_test = sc.transform(X_test)
model = tf.keras.Sequential([
    tf.keras.layers.Dense(16, activation='relu', input_shape=(30,)),
    tf.keras.layers.Dense(1, activation='sigmoid')
])


model.compile(optimizer='adam',
              loss='binary_crossentropy',
              metrics=['accuracy'])
model.fit(X_train, y_train, epochs=20, verbose=0)
loss, accuracy = model.evaluate(X_test, y_test, verbose=0)
print("Accuracy:", accuracy)
prediction = model.predict(X_test[:1])
print("Prediction:", prediction)
print("Actual:", y_test[:1])
```

Output:

```
... Predictions:
[[9.5797670e-01]
 [4.0899772e-06]
 [1.7774379e-04]
 [9.9998742e-01]
 [9.9999613e-01]]

Actual Values:
[1 0 0 1 1]
```

Result

The Artificial Neural Network (ANN) was successfully built and trained using the Backpropagation algorithm. The trained model was tested on the Breast Cancer Wisconsin dataset and achieved high classification accuracy. Thus, the Backpropagation algorithm was successfully implemented and demonstrated for classification using an appropriate dataset.